

devices and systems. Arduino is a firmware platform catering to the needs of DIY projects. Node-RED is a visual programming tool that is useful for creating workflows for IoT applications.

Augmented reality (AR)

AR focuses on overlaying digital information of the real world on the user's view. It found its way into game development initially, and has today spread across various industries and use cases. ARToolKit and ARCore are libraries for building AR applications on Windows, Linux, and iOS. Its ability to track and overlay digital content on real-world objects has been emerging into many everyday use cases in airports, shopping malls, and other venues. OpenCV and Vuforia are platforms that provide a wide range of image processing and object recognition abilities.

Virtual reality (VR)

VR supports the creation of a computer-generated, multi-dimensional interactive environment with the help of special equipment like a VR headset. Many commercial attempts have failed to capitalise on this technology -- from Google to Apple to recent Meta platforms. However, the open source community is thriving and developing new innovative ways in which it can be used. OpenVR and WebVR are at the forefront of driving the development of VR applications. A-Frame has become the popular framework for building AR-integrated VR experiences using HTML, CSS, and JavaScript; it's built on WebGL. OpenSim is a multi-platform for 3D simulation and visualisation allowing users to create their own virtual worlds; it supports Oculus and Vive.

Robotics

Robotics aims to create intelligent machines that can assist us in various ways. With Intrinsic acquiring ROS (Robot Operating System), a popular

open source platform that drives the development of autonomous robots and systems, one can hope for the best year ahead with new innovative solutions coming to support the open source community. OpenCV is another similar platform for robotics development with a rich vision library that can be used to detect and track objects, faces and more. Gazebo is simulation software that allows the simulation and testing of robot software in a realistic environment. MoveIt is popular in motion planning capabilities for various robotic arms and mobile bases.

3D printing

3D printing has advanced beyond the manufacturing industry. RepRap and OctoPrint lead the development of 3D printing applications. Cura and PrusaSlicer are slicing software developed with rich features for handling 3D printers.

Drones

Drones are flying robots that can be remotely controlled or fly autonomously using software-controlled flight plans with onboard sensors and GPS. PX4 and ArduPilot are popular platforms for the development of autonomous drones.

Paparazzi UAV is an autopilot system for fixed-wing and multi-copter drones used for DIY and model development purposes. MIT-UAV is research-grade technology used in complex autonomous aircraft use cases.

Biotechnology

Biotechnology addresses the medical, agricultural, industrial, food, and marine industries in one broad stroke. Genetics engineering has been its focus area, and helped to develop the first drug for overcoming the Corona virus. OpenPCR and OpenTrons are popular development tools used for biological and medical applications. OpenWetWare is a platform that allows sharing of protocols, data, and information about research projects. DIYbio exposes citizens to biology projects, and is a promising and innovative open source platform.

As the year unfolds, one thing is certain: technology is going to play a major role in every facet of our lives and innovation is going to continue making everyday life easier. Talent shortage is one of the top concerns IT leaders are faced with when adopting emerging technologies. However, open source platforms thrive better than others, and help groom next-generation talent. **END** 🐧

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